

# School of Natural Sciences and Mathematics

## Mathematics (BA, BS)

Mathematics is both a profession and an indispensable tool for many types of work. As a tool, mathematics is a universal language that has been crucial in formulating and expressing ideas not only in science and engineering, but also in many other areas such as business and the social sciences. As probably the oldest and most basic science, it provides the key to understanding the major technological achievements of our time.

Of equal importance, knowledge of mathematics may help provide a student with the type of uncompromising and clear-sighted thinking useful in considering the problems of many other disciplines. The Mathematics degree program encompasses mathematics, statistics, and applied mathematics. The Mathematical Sciences Department also administers a [Bachelor of Science in Actuarial Science](#).

Students interested in obtaining either a BA or a BS in Mathematics and Teacher Certification in the state of Texas should consult the [UTeach Dallas office](#) or the [Teacher Development Center](#) for specific requirements as soon as possible after formal admission to the University. See the Teacher Education Certification Programs section of the catalog for additional information.

The Mathematics degree program also prepares students for graduate studies. [Accelerated BS/MS Fast Track programs](#) are available that provide the opportunity to partially satisfy the requirements of a master's degree while completing the bachelor's degree in Mathematics.

## The Program in Mathematics

Students seeking a degree in Mathematics may choose from a variety of courses, which allow for flexibility and specialization so that students can better adapt their educational experience to further their academic goals. Students entering the field of mathematics at UT Dallas are prepared not only for careers in these areas, but also in numerous other professions that require expertise in mathematics.

Students who pursue a BS degree in mathematics typically use their degree in the fields of mathematics, statistics, science, or engineering. BS students choose a specialization in either mathematics, statistics or applied mathematics.

**Mathematics Specialization:** For students interested in a career in mathematics and for those who choose to continue their education at the graduate level in mathematics, applied mathematics, mathematics education or disciplines, such as statistics and data science, which use mathematics.

Statistics Specialization: For students interested in probability and statistical models and their use in data analysis and decision making; for students interested in continuing on to graduate work in statistics, biostatistics, data science, actuarial science, and other statistics-related areas.

Applied Mathematics Specialization: For students interested in mathematics for the purpose of using it broadly in various application areas and for students interested in continuing on to graduate work in applied mathematics or related areas.

Students who pursue a BA degree in mathematics generally use mathematics as background for study in fields such as arts and technology, education, engineering or science.

## Bachelor of Arts in Mathematics

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

All majors are strongly urged to meet with assigned departmental advisors every semester

## Faculty

FACG> nsm-mathematics-ba

**Professors:** Swati Biswas, Min Chen, Pankaj Choudhary, Baris Coskunuzer, Mieczyslaw Dabkowski, Vladimir Dragovic, Sam Efromovich, Wieslaw Krawcewicz, Susan Minkoff, L. Felipe Pereira, Dmitry Rachinskiy, Viswanath Ramakrishna, Janos Turi

**Associate Professors:** Maxim Arnold, Yan Cao, Liang Hong, Oleg Makarenkov, Tomoki Ohsawa, Anh Tran

**Assistant Professors:** Carlos Arreche, Noirrit Chandra, Ronan Conlon, Rizwanur Khan, Qiwei Li, Stephen McKeown, Chuan-Fa Tang, Jiayi Wang, Nathan Williams, Nan Wu

**Professors Emeriti:** Larry Ammann, Ali Hooshyar, Patrick Odell, John Van Ness

**Clinical Professor:** Natalia Humphreys

**Clinical Associate Professor:** Mohammad Akbar

**Clinical Assistant Professor:** Wenyi Lu

**Professors of Instruction:** Anatoly Eydelzon, Manjula Foley, Bentley Garrett, Yuly Koshevnik

**Associate Professors of Instruction:** Mohammad Ahsan, Kelly Aman, Malgorzata Dabkowski, Rabin Dahal, Derege Mussa, My Linh Nguyen, Jigarkumar Patel, Julie Sutton

**Assistant Professors of Instruction:** Anani Komla Adabrah, Iris Alvarado, Saikat Biswas, Hui Ding, Kemelli Estacio-Hiroms, Huizhen Guo, Shengjie Jiang, Joselle Kehoe, Runzhou Liu, Neha Makhijani, Irina Martynova, Diarisoa Mihaja Rakotomalala, Adrian Murza, Ajaya Paudel, Octavious Smiley, Nasrin Sultana, Che-Yu Wu

**I. Core Curriculum Requirements: 42 semester credit hours**

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

**Communication: 6 semester credit hours**

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

**Mathematics: 3 semester credit hours**

[MATH 2417](#) Calculus I <sup>1, 2, 3</sup>

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

**Life and Physical Sciences: 6 semester credit hours** <sup>1, 4</sup>

[PHYS 2325](#) Mechanics

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat

[PHYS 2326](#) Electromagnetism and Waves

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

**Language, Philosophy and Culture: 3 semester credit hours**

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

**Creative Arts: 3 semester credit hours**

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

**American History: 6 semester credit hours**

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

**Government/Political Science: 6 semester credit hours**

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

**Social and Behavioral Sciences: 3 semester credit hours**

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

**Component Area Option: 6 semester credit hours**

[MATH 2417](#) Calculus I <sup>1, 2, 3</sup>  
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[MATH 2419](#) Calculus II <sup>1, 2, 3</sup>  
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[PHYS 2125](#) Physics Laboratory I <sup>1, 4</sup>  
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or [PHYS 2421](#) Honors Physics I - Mechanics and Heat <sup>5, 6</sup>  
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Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 49-50 semester credit hours

**Major Preparatory Courses: 16-17 semester credit hours beyond Core Curriculum**

[NATS 1101](#) Natural Sciences and Mathematics Freshman Seminar

[PHYS 2325](#) Mechanics <sup>1, 4</sup>  
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and [PHYS 2125](#) Physics Laboratory I <sup>1, 4</sup>  
\_ \_

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat <sup>1, 6, 7</sup>  
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[PHYS 2326](#) Electromagnetism and Waves <sup>1, 4</sup>  
\_ \_

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves <sup>1, 4</sup>  
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[PHYS 2126](#) Physics Laboratory II

[MATH 2306](#) Analytic Geometry

[MATH 2370](#) Introduction to Programming with MATLAB

or [CS 1325](#) Introduction to Programming <sup>8</sup>  
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or [CS 1337](#) Computer Science I <sup>8</sup>  
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[MATH 2417](#) Calculus I <sup>1, 2, 3</sup>  
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[MATH 2418](#) Linear Algebra<sup>8</sup>

[MATH 2419](#) Calculus II<sup>1, 2, 3</sup>

[MATH 2420](#) Differential Equations with Applications<sup>8</sup>

**Major Core Courses: 24 semester credit hours**

[MATH 3310](#) Theoretical Concepts of Calculus

[MATH 3311](#) Abstract Algebra I

[MATH 3380](#) Differential Geometry

[MATH 3323](#) Elementary Number Theory

[MATH 3379](#) Complex Variables

[MATH 3351](#) Advanced Calculus

[STAT 4351](#) Probability

[STAT 4352](#) Mathematical Statistics

**Major Related Courses: 9 semester credit hours**

Nine semester credit hours of upper-division MATH, STAT or ACTS courses, at least six of which must be MATH courses at the 4000-level. These courses cannot include those for which the catalog entry states: May not be used to satisfy mathematics requirements by students in Mathematics.

**III. Elective Requirements: 28-29 semester credit hours**

**Electives: 28-29 semester credit hours**

Both lower- and upper-division courses may count as electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

1. A required Major course that also fulfills Core Curriculum requirements. If semester credit hours are counted in the Core Curriculum, students must complete additional coursework to meet the minimum requirement for graduation. Course selection assistance is available from the undergraduate advisor.

2. Three semester credit hours of Calculus are counted to fulfill the Mathematics Core Requirement with the remaining one semester credit hour to be counted under Component Area Option Core.

3. MATH 2417 and MATH 2419 requirements can be fulfilled by completing MATH 2413, MATH 2414, and MATH 2415.
4. Six semester credit hours of Physics are counted under Science core, and one semester credit hour of Physics (PHYS 2125) is counted under Component Area Core.
5. Please consult your advisor if selecting Honors Physics.
6. Students may use three semester credit hours of PHYS 2421 to count under Science core, and one semester credit hour of PHYS 2421 under Component Area Option core.
7. Students who complete PHYS 2421 do not need to complete PHYS 2125.
8. Indicates a prerequisite class to be completed before enrolling in upper-division classes.

## Bachelor of Science in Mathematics

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

All majors with specialization in either Mathematics, Statistics, or Applied Mathematics are strongly urged to meet with the assigned undergraduate advisor from the Mathematical Sciences Department every semester.

## Faculty

FACG> nsm-mathematics-bs

**Professors:** Swati Biswas, Min Chen, Pankaj Choudhary, Baris Coskunuzer, Mieczyslaw Dabkowski, Vladimir Dragovic, Sam Efromovich, Wieslaw Krawcewicz, Susan Minkoff, L. Felipe Pereira, Dmitry Rachinskiy, Viswanath Ramakrishna, Janos Turi

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**Clinical Professor:** Natalia Humphreys

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I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

**Communication: 6 semester credit hours**

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

**Mathematics: 3 semester credit hours**

[MATH 2417](#) Calculus I [9](#), [10](#)

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

**Life and Physical Sciences: 6 semester credit hours** [9](#), [11](#)

*Mathematics/Applied Mathematics Specialization*

[PHYS 2325](#) Mechanics

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat

[PHYS 2326](#) Electromagnetism and Waves

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves

*Statistics Specialization*

[PHYS 2325](#) Mechanics

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat

[PHYS 2326](#) Electromagnetism and Waves

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves

Or

[CHEM 1311](#) General Chemistry I

or [CHEM 1315](#) Honors Freshman Chemistry I

[CHEM 1312](#) General Chemistry II

or [CHEM 1316](#) Honors Freshman Chemistry II

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

**Language, Philosophy and Culture: 3 semester credit hours**

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

**Creative Arts: 3 semester credit hours**

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

**American History: 6 semester credit hours**

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

**Government/Political Science: 6 semester credit hours**

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

**Social and Behavioral Sciences: 3 semester credit hours**

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

**Component Area Option: 6 semester credit hours**

[MATH 2417](#) Calculus I <sup>9, 10</sup>

[MATH 2419](#) Calculus II <sup>9, 10</sup>

[PHYS 2125](#) Physics Laboratory I <sup>9, 11</sup>

or [CHEM 1111](#) General Chemistry Laboratory I <sup>9, 11</sup>

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat <sup>12, 13</sup>

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)



II. Major Requirements: 49-50 semester credit hours

**Major Preparatory Courses: 13-14 semester credit hours beyond Core Curriculum**

[NATS 1101](#) Natural Sciences and Mathematics Freshman Seminar

*For Mathematics/Applied Mathematics Specialization*

[PHYS 2325](#) Mechanics [9](#), [11](#)

and [PHYS 2125](#) Physics Laboratory I [9](#), [11](#)

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat [9](#), [13](#), [14](#)

[PHYS 2326](#) Electromagnetism and Waves [9](#), [11](#)

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves [9](#), [11](#)

[PHYS 2126](#) Physics Laboratory II

*For Statistics Specialization*

[PHYS 2325](#) Mechanics [9](#), [11](#)

and [PHYS 2125](#) Physics Laboratory I [9](#), [11](#)

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat [9](#), [13](#), [14](#)

[PHYS 2326](#) Electromagnetism and Waves [9](#), [11](#)

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves [9](#), [11](#)

[PHYS 2126](#) Physics Laboratory II

Or

[CHEM 1111](#) General Chemistry Laboratory I [9](#), [11](#)

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 1311](#) General Chemistry I [9](#), [11](#)

or [CHEM 1315](#) Honors Freshman Chemistry I [9](#), [11](#)

[CHEM 1312](#) General Chemistry II [9](#), [11](#)

or [CHEM 1316](#) Honors Freshman Chemistry II [9](#), [11](#)

*For All*

[MATH 2370](#) Introduction to Programming with MATLAB

or [CS 1325](#) Introduction to Programming [15](#)

or [CS 1337](#) Computer Science I [15](#)

[MATH 2417](#) Calculus I [9](#), [10](#), [16](#)

[MATH 2418](#) Linear Algebra [15](#)

[MATH 2419](#) Calculus II [9](#), [10](#), [16](#)

[MATH 2420](#) Differential Equations with Applications [15](#)

### **Major Core Courses: 24 semester credit hours**

[MATH 3310](#) Theoretical Concepts of Calculus

[MATH 3311](#) Abstract Algebra I

[MATH 3379](#) Complex Variables

[MATH 3351](#) Advanced Calculus

[MATH 4301](#) Mathematical Analysis I

[MATH 4302](#) Mathematical Analysis II

[MATH 4334](#) Numerical Analysis

[STAT 4351](#) Probability

### **Major Related Courses: 12 semester credit hours**

*Applied Mathematics Specialization*

[MATH 4341](#) Topology

[MATH 4355](#) Methods of Applied Mathematics

[MATH 4362](#) Partial Differential Equations

[STAT 4382](#) Stochastic Processes

*Mathematics Specialization*

[MATH 3312](#) Abstract Algebra II

[MATH 3380](#) Differential Geometry

[MATH 4341](#) Topology

3 semester credit hours upper-division guided elective<sup>17</sup>

*Statistics Specialization*

[STAT 3355](#) Introduction to Data Analysis

[STAT 4352](#) Mathematical Statistics

[STAT 4382](#) Stochastic Processes

3 semester credit hours upper-division guided elective<sup>17</sup>

III. Elective Requirements: 28-29 semester credit hours

**Electives: 28-29 semester credit hours**

Both lower- and upper-division courses may count as electives

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

9. A required Major course that also fulfills Core Curriculum requirements. If semester credit hours are counted in the Core Curriculum, students must complete additional coursework to meet the minimum requirement for graduation. Course selection assistance is available from the undergraduate advisor.

10. Three semester credit hours of Calculus are counted to fulfill the Mathematics Core Requirement with the remaining one semester credit hour to be counted under Component Area Option Core.

11. Six semester credit hours of Physics or Chemistry are counted under Science core, and one semester credit hour of Physics or Chemistry (PHYS 2125 or CHEM 1111) is counted under Component Area Core.

12. Please consult your advisor if selecting Honors Physics.

13. Students may use three semester credit hours of PHYS 2421 to count under Science core, and one semester credit hour of PHYS 2421 under Component Area Option core.

14. Students who complete PHYS 2421 do not need to complete PHYS 2125.

15. Indicates a prerequisite class to be completed before enrolling in upper-division classes.

16. MATH 2417 and MATH 2419 requirements can be fulfilled by completing MATH 2413, MATH 2414, and MATH 2415.

17. Approval of Mathematics department advisor required.

NOTE> Please be advised, the UTeach section below feeds in from a separate page updated by the UTeach team. Any changes made to the UTeach section below will not be retained.

## UTeach Option

The [UTeach option](#) may be added to either the BA or BS degree. UTeach Dallas Option degree plans are streamlined to allow students to complete both a rigorous Bachelor of Science or Bachelor of Arts degree and all coursework for middle or high school teacher certification in four years. Teaching Option degrees require deep content knowledge combined with courses grounded in the latest research on math and science education. While most graduates go on to classroom teaching, UTeach alums are also prepared to enter graduate school and to work in a discipline related industry.

## Fast Track Baccalaureate/Master's Degrees

For students with strong academic record who intend to pursue master's studies at UT Dallas, accelerated BS/MS Fast Tracks are available. After Fast Track admission to an MS program, students may take up to 15 semester credit hours of approved graduate courses in their senior year to use toward completion of both the BS and MS degrees. The available Fast Track options and their admission requirements, if any, in addition to having completed 90 or more semester credit hours (out of which 36 hours are from the core curriculum) with a cumulative GPA of at least 3.00, are as follows:

- [MS in Mathematics](#): Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.
- [MS in Statistics](#): Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.

Interested students, after reviewing the [UT Dallas Fast Track policy](#), should contact their undergraduate advisor and the graduate advisor of the intended MS program well in advance of their junior year to prepare a course sequence permitting maximal advantage and apply to the

Fast Track program.

## Minors

Students must take a minimum of 18 semester credit hours for the minor, 12 of which must be upper-division semester credit hours. Students who take a minor will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average). Semester credit hours may not be used to satisfy both the major and minor requirements; however, free elective semester credit hours or major preparatory classes may be used to satisfy the minor.

For all minors in the School of Natural Sciences and Mathematics students must complete all prerequisite sequences for required minor courses.

### Minor in Mathematics

18 semester credit hours

The minor in Mathematics requires 18 semester credit hours math or statistics course requirements. Of these 18, 12 semester credit hours will be selected from a specific set of courses.

**12 semester credit hours of courses must be chosen from the following:**

[MATH 3310](#) Theoretical Concepts of Calculus

[MATH 4334](#) Numerical Analysis

And select two more upper-division mathematics courses that satisfy degree requirements by students in Mathematical Sciences.

The remaining 6 semester credit hours can be satisfied by choosing either MATH or STAT courses with advisor approval.

### Minor in Statistics

18 semester credit hours

The minor in Statistics requires 18 semester credit hours math or statistics course requirements. Of these 18, 12 semester credit hours will be selected from a specific set of courses.

**12 semester credit hours of courses must be chosen from the following:**

[STAT 4351](#) Probability

## [STAT 4352](#) Mathematical Statistics

And select two more upper-division mathematics courses that satisfy degree requirements by students in Mathematical Sciences.

The remaining 6 semester credit hours can be satisfied by choosing either MATH or STAT courses with advisor approval.

## Certificates

### Certificate in Biomedical Data Analytics

15 semester credit hours

A Certificate in Biomedical Data Analytics is offered by the Department of Mathematics in the School of Natural Sciences and Mathematics.

The Certificate in Biomedical Data Analytics covers programming, bioinformatics, and biostatistical and machine learning methods with application to statistical analysis of data commonly arising in biomedical, epidemiological, and healthcare related sciences.

#### Admission Requirements

Open to both current UT Dallas students and individuals who are external to UT Dallas.

Required prerequisite of two semesters of Calculus:

[MATH 2413](#) Differential Calculus

[MATH 2414](#) Integral Calculus

#### Certificate Requirements

Students must complete the following courses over 4 semesters:

[STAT 2332](#) Introductory Statistics for Life Sciences

or [STAT 3332](#) Statistics for Life Sciences

[BIMS 3335](#) Informatics and Programming

or [MATH 3335](#) Informatics and Programming

or [STAT 3335](#) Informatics and Programming

[BIMS 3336](#) Bioinformatics

or [MATH 3336](#) Bioinformatics

or [STAT 3336](#) Bioinformatics

or [BIOL 3337](#) Bioinformatics

[BIMS 3337](#) Elements of Biostatistics and Epidemiology

or [STAT 3337](#) Elements of Biostatistics and Epidemiology

[BIMS 4338](#) Biostatistics and Machine Learning Lab

or [STAT 4338](#) Biostatistics and Machine Learning Lab

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